Sanaul Haque

Rojamin_Rahman_Proposal

Course 201

Document Details

Submission ID
trn:oid:::3117:636712647

Submission Date
Oct 10, 2026, 12:05 AM GMT+6

Download Date
Oct 10, 2026, 12:07 AM GMT+6

File Name
Rojamin_Rahman_Proposal.pdf

File Size
355.9 KB

8 Pages

2,538 Words

14,693 Characters

32% Overall Similarity

The combined total of all matches, including overlapping sources, for each database.

Match Groups

- **31** Not Cited or Quoted 27%
  Matches with neither in-text citation nor quotation marks
- **4** Missing Quotations 1%
  Matches that are still very similar to source material
- **4** Missing Citation 4%
  Matches that have quotation marks, but no in-text citation
- **0** Cited and Quoted 0%
  Matches with in-text citation present, but no quotation marks

Top Sources

29% Internet sources
26% Publications
27% Submitted works (Student Papers)

Match Groups

- **31** Not Cited or Quoted 27%
  Matches with neither in-text citation nor quotation marks
- **4** Missing Quotations 1%
  Matches that are still very similar to source material
- **4** Missing Citation 4%
  Matches that have quotation marks, but no in-text citation
- **0** Cited and Quoted 0%
  Matches with in-text citation present, but no quotation marks

Top Sources

29% Internet sources
26% Publications
27% Submitted works (Student Papers)

Top Sources

The sources with the highest number of matches within the submission. Overlapping sources will not be displayed.

1	Internet	arxiv.org 2%
2	Student papers	BITS, Pilani-Dubai on 2024-11-27 2%
3	Student papers	Hong Kong University of Science and Technology on 2025-05-10 2%
4	Publication	Nicole Arias Espinoza, Stefany Solano González. "In Silico Identification of Potenti... 2%
5	Publication	Sambhaji Vyavahare, Atharva Pawar, Shafiya A. Sayyed, Neha Shegar, Sakshi Ray... 2%
6	Internet	core.ac.uk 2%
7	Internet	kuet.ac.bd 1%
8	Internet	tpmap.org 1%
9	Internet	www.mrforum.com 1%
10	Student papers	Nerinx Hall High School on 2025-05-15 1%

11	Internet	
cris.maastrichtuniversity.nl		1%
12	Internet	
iarjset.com		1%
13	Student papers	
Monash University on 2020-10-21		1%
14	Internet	
ouci.dntb.gov.ua		1%
15	Student papers	
University of the West Indies on 2026-04-11		1%
16	Publication	
Yuushou Nakayama, Tomoka Shuto, Ryo Tanaka, Takeshi Shiono, Shodai Hino, No...		1%
17	Publication	
Quyuan Xiong, Guangyu Liu, Shipeng Gao, Yuechuo Hao et al. "Artificial Intelligen...		1%
18	Student papers	
California State University, Sacramento on 2023-05-14		<1%
19	Internet	
www.zgr.net		<1%
20	Student papers	
Monash University on 2026-09-19		<1%
21	Student papers	
Birla Institute of Technology and Science Pilani on 2021-08-09		<1%
22	Student papers	
University of Durham on 2024-01-19		<1%
23	Internet	
sy.bilpubgroup.com		<1%
24	Student papers	
University of Stellenbosch, South Africa on 2023-04-28		<1%

25 Internet

www.science.org <1%

26 Internet

www.wseas.com <1%

27 Publication

Neha Saini, Diksha Bharti, Shivani Singh, Ruksana Banoo, Vindhya Bundela, Pooja... <1%

28 Publication

Saba Yousafzai, Mujahid Farid, Muhammad Zubair, Nafeesa Naeem, Wardha Zafa... <1%

29 Internet

jitcai.org <1%

30 Student papers

University System of Georgia on 2025-11-21 <1%

31 Internet

profilpelajar.com <1%

32 Internet

revistas.unal.edu.co <1%

33 Internet

www.biotech-asia.org <1%

INTERNATIONAL ACADEMIC RESEARCH COMPETITION 2026

RESEARCH PROPOSAL

Full Legal Name: Rojamin Rahman

Institution: Dhaka City College

Category: Junior

Grade: Grade 12

Country: Bangladesh

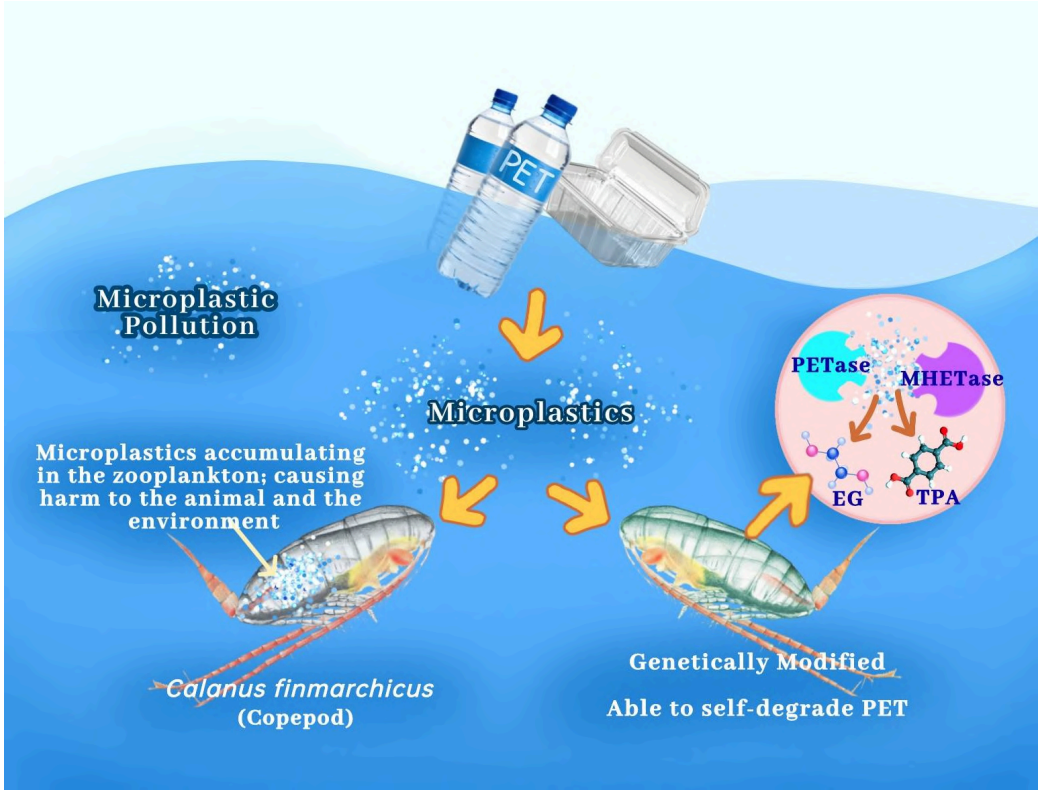
Submission Date: September 19, 2026

Registered Email Address: rojaminrahman@gmail.com

Research Topic: Genetic Modification of Copepods with Plastic-degrading Enzyme Genes for Tackling Microplastic Pollution in Oceanic Environment

Title: Genetic Modification of Copepods with Plastic-degrading Enzyme Genes for Tackling Microplastic Pollution in Oceanic Environment

Graphical Abstract



Ocean zooplankton suffer from the microplastic pollution which is detrimental to both its well-being and the environment. However, genetically modified zooplankton can tackle the issue itself by being able to break down PET to its building molecules.

Introduction

The oceans are one of the most crucial elements in our biosystem, which continues to make the earth habitable. Planktons, which are tiny living organisms found in waters, carry out important processes that maintain the marine ecosystem's functionality and balance. Zooplankton, the animal plankton of the sea, plays significant roles in oceanic food and nutrient cycle as well as ocean carbon sequestration [1]. Among them, copepods are found universally in all oceans. Because of the rise of plastic wastes everywhere, it is causing another alarming issue which is microplastic pollution, the exposure of tiny plastic particles to any system or environment [2]. Microplastic pollution is high near coastal areas due to human activities and impacts the marine environment terribly [3]-[4]. One of the problems it is responsible for is that microplastics can be ingested by copepods as the tiny particles can be mistaken as food for them [5]. This disrupts copepods' biological activities, including the processes which are essential for natural bioremediation [6]. Moreover, copepods are consumed by higher level consumers in the food chain, meaning the microplastics accumulated in the zooplankton can affect the health of other

animals, including humans [7]. In recent studies, a plastic-degrading bacterium, *Ideonella sakaiensis*, has been found [8]. This bacterium can produce PETase and MHETase enzymes, which can degrade polyethylene terephthalate (PET), the plastic polymer found widely in plastic bottles and containers [9]. Previously, scientists were successful in their approach to enabling bacteria, phytoplankton and microalgae to be genetically engineered to produce the plastic-degrading enzymes [10]-[13]. Therefore, this research aims to genetically modify copepods with genes responsible for PETase and MHETase enzymes by utilizing microinjection techniques on the embryonic level. This will enable the transformed copepods to self-degrade the microplastics they consume.

Research Questions

This research will aim to answer the following questions:

1. Can copepods be successfully genetically engineered with PETase and MHETase producing genes in the embryonic stage applying microinjection techniques?
2. Can the transgenic copepods secrete PETase and MHETase enzymes when coming in contact with microplastics? If not, what are the factors blocking gene expression?
3. What is the scope of the transgenic copepods' microplastic degradation on different parameters?
4. Are the transgenic copepods able to adapt to the natural biosystem?

Hypotheses

This research will aim to test out the following hypotheses:

1. Null Hypothesis (H_0): It is not possible to genetically modify copepods with the plastic-degrading enzyme genes in the embryonic stage using microinjection, and they cannot degrade any plastic particles.
2. Alternate hypothesis (H_1): It is possible to genetically modify copepods with the plastic-degrading enzyme genes in the embryonic stage using microinjection, but they cannot degrade microplastics for flawed gene expression.
3. Alternate hypothesis (H_2): It is possible to genetically modify copepods with the plastic-degrading enzyme genes in the embryonic stage using microinjection, and they can break down plastic particles to some extent by producing the necessary enzymes.

Literature Review

Zooplankton are significant to the marine biome and the atmosphere as they partake into sinking carbon dioxide to deep waters from the air, recycling inorganic nutrients back to producers, controlling the phytoplankton population, shifting energy to higher trophic creatures and many more [1]. They play a significant role in food chains; small fishes feed on them, then they are consumed by larger fishes. Therefore, the health of zooplankton are important to the animals and human populace as fish and other ocean beings are consumed by them [7].

Microplastic pollution is a critical issue all over the world and scientists predict that microplastics will continue to increase throughout the future decades [4]. From the ocean's standpoint, it poses a great threat to coastal areas and living creatures [3]. One of the conundrums it creates is that it harms the zooplankton by entering in their stomach [5]. As a result, these creatures produce less energy, which affects their biological cycles including reproduction, growth and lifespan [6]. This also risks microplastic transfer to animals including humans as zooplankton are considerably integrated in the ocean food web [7].

Ideonella sakaiensis, a rod-shaped gram-negative bacterium, is able to break down plastics by producing enzymes called PETase and MHETase [8]. PETase can reduce PET plastic to mono-(2-hydroxyethyl) terephthalate (MHET), and MHETase can simplify MHET to ethylene glycol (EG) and terephthalic acid (TPA) – the building blocks of PET [14].

Prior studies have shown successful genetic modification in bacteria: *Escherichia coli*, *Bacillus subtilis* and microalgae: *Phaeodactylum tricornutum*, *Chlamydomonas reinhardtii* with PETase enzyme and being able to degrade plastics [10]-[13]. There remains potential for making zooplankton such as copepods to be enabled to degrade plastic with *Is*PETase (the PETase enzyme found in *I. sakaiensis*) with possible host optimization opportunities to the *Is*PETase gene [15]-[16].

As it can be observed, zooplankton are yet to be approached for successful genetic transformation with plastic-degrading enzymes and modified to degrade microplastics. Allowing copepods to self-degrade microplastics themselves will aid in undertaking microplastic pollution in the ocean, its effects on copepods and other animals. This study will utilize microinjection to transfer the plastic-degrading enzyme genes in the embryonic phase of copepod, *Calanus finmarchicus*, to enable them to break down microplastics.

Proposed Methodology

- **Species cultivation:** The target copepod species for this research is *Calanus finmarchicus*. Mature pairs of *C. finmarchicus* will be collected and cultivated in a controlled environment with necessary growth media (e.g. nutrients) and environmental factors (e.g. temperature, water salinity).
- **Gene optimization and transmission:** Genes for PETase (Gene code: ISF6-4831) and MHETase (Gene code: ISF6-0224) will be refined to suit the host by optimizing the DNA sequence. Newly spawned single-celled eggs will be electroporated to allow the optimized DNA sequences of the genes to be injected into the egg cell with microinjection.
- **DNA verification:** After the injected batch has reached adulthood, samples of them will be collected to verify the successful transmission of the genes into their DNA. To achieve this, the cell samples will be centrifuged in a microtube. The DNA will be separated from the pellet through gel electrophoresis. The enzyme gene sequences can be identified with DNA sequencing.
- **Evaluate plastic degradation:** The microplastic degradation ability of the transgenic *C. finmarchicus* will be tested. Samples will have a specific, innocuous amount of PET particles supplied and later checked if they are able to degrade plastic using Raman

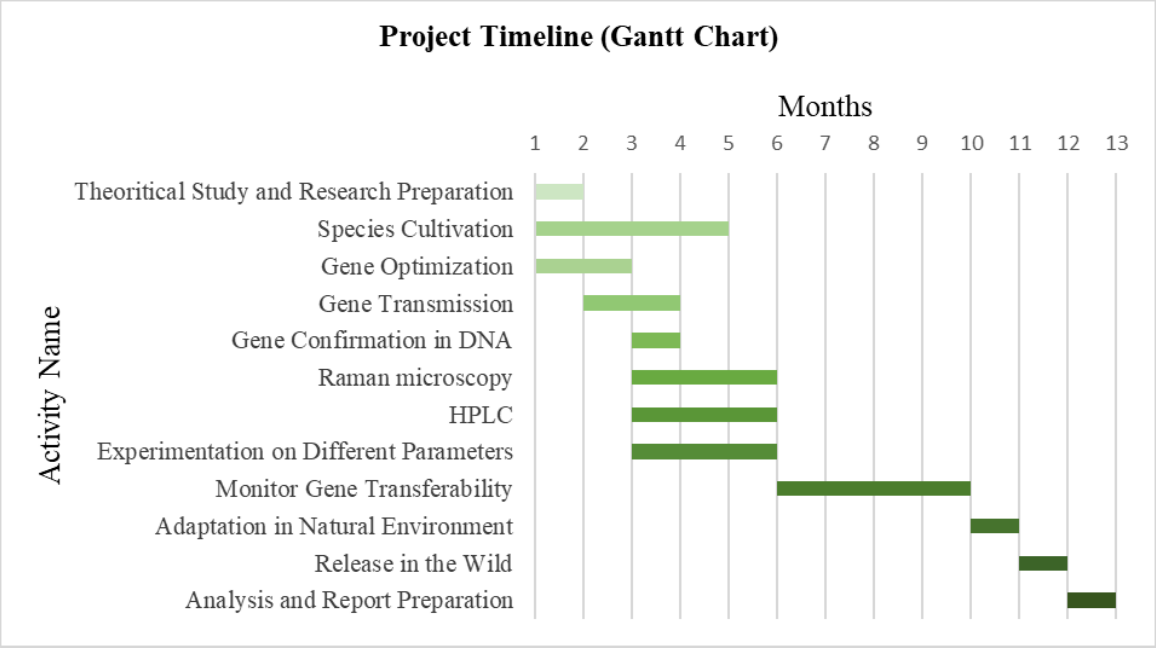
turnitin Page 9 of 13 - Integrity Submission Submission ID trn:oid:::3117:636712647

microscopy. Plastic degradation will be analyzed with high-performance liquid chromatography (HPLC) and observed on different parameters such as temperature and pH level.

- **Monitor gene transferability:** The transformed target species will be cultivated on a large scale. Reproduction between a transgenic pair and one normal, one transgenic pair of *C. finmarchicus* will be observed to understand the gene transmission in new generations.
- **Adaptation and release:** The transformed *C. finmarchicus* will be discharged into natural water bodies. After confirming their positive adaptation, they will be released in the wild.

Project Timeline

This project is envisioned to be completed within a timeframe of 12 months. The Gantt chart of the project timeline illustrating the different tasks and time assigned for them (in months) is shown below:



Project Practicalities

- **Appropriate target species:** *C. finmarchicus* is a highly adaptable copepod and found almost universally in oceans. It makes them the most suitable host for the project's success as well as worldwide application.
- **Timeframe:** This project is achievable within the proposed timeline.
- **Practicable methods:** Microinjection is a viable method to transfer genetic material into the proposed host of the research and for genetic engineering. This technique has been utilized in similar research before [17]-[18].

- **Diverse plastic degradation range:** PETase plastic degradation is not limited to PET plastic only. It has shown to break down plastics with similar ester structures.
- **Feasible results:** Similar experiments were achieved effectively before on different creatures with favorable results.

Potential Limitations

- Some potential challenges of the proposed project are: the necessity of a sizable, proper lab facility and the high costing of the materials and equipment for lengthy use, which requires the acquisition of sufficient funding.
- The research has a lengthy duration for multiple experiment trials for precise findings, analysis and conclusion. Large scale execution of the project also poses difficulty with time and resources.
- PETase plastic degradation capabilities are restricted to similar ester-like plastics. It cannot degrade other common plastics such as polyethylene. Also, TPA from degraded PET could be toxic to the target species.

Expected Outcome

We expect to successfully modify the copepod, *C. finmarchicus*, to acquire the ability to self-degrade the microplastics they ingest by secreting PETase and MHETase when coming in contact with PET microplastics. This can prevent the hindrance in energy production of the zooplankton, which is detrimental to their biological operations, while also reducing microplastic contamination in food cycles and in the ocean.

Conclusion

Genetically modified copepods with plastic-degrading capabilities are promising for the goal of diminishing microplastic pollution in the ocean. The suggested approach has its merits in being environmentally friendly and dependable for tackling ocean microplastic pollution and the harmful effect it has on ocean dwellers and other living beings. Different living creatures that were previously engineered by scientists with similar goals have shown positive results. The selected framework for this project is simple and trustworthy. Research results from this study can prove to become a pillar for future advancement in environment restoration. Further improved techniques and widespread implementation is hopeful for a sustainable future.

References

15 7 [1] P. Chowdhury, S. J. Khan, Md Afsar Ahmed Sumon, Mohamed Ahmed Abu El-Regal, K. Tayyamath, and H. V. Doan, "The role of planktons in sustaining the function and diversity of marine ecosystems: a systematic review," *Discover Animals*, vol. 3, no. 9, pp. 1-20, 2026, doi: https://doi.org/10.1007/s44338-025-00113-7

1 [2] J. R. Bermúdez, M. Metian, P. W. Swarzenski, M. S. Bank, R. Bjorøy, J. Cajas, R. Bucheli, R. González-Muñoz, J. Lynch, E. Piguave, N. Vargas, K. Vilela, L. Calle, M. J. Borbor-Cordova, and N. Gaibor, "Marine microplastics on the rise in the Eastern Tropical Pacific: Abundance

doubles in 11 years and a ten-fold increase is projected by 2100," *Marine Pollution Bulletin*, vol. 221, pp. 118437, 2025, doi: https://doi.org/10.1016/j.marpolbul.2025.118437

[3] G. Y. Gaviño-Romero, Alma Paola Rodríguez Troncoso, L. C. Alarcón-Ortega, E. Bautista-Guerrero, and A. L. Cupul-Magaña, "Potential microplastic pollution: an invisible threat to the success of coral reef restoration efforts," *International Journal of Tropical Biology and Conservation*, vol. 74, pp. 1-15, 2026, doi: https://doi.org/10.15517/pq3hka26

[4] O. Ahmad, M. Jamal, H. Almalki, A. Alzahrani, A. Alatawi, and M. Haque, "Microplastic pollution in the marine environment: Sources, impacts, and degradation.," *Journal of Advanced Veterinary and Animal Research*, vol. 12, no. 1, pp. 260-279, 2025, doi: http://doi.org/10.5455/javar.2025.l893

[5] V. S. Lin, "Research highlights: impacts of microplastics on plankton," *Environmental Science: Processes & Impacts*, vol. 18, no. 2, pp. 160-163, 2016, doi: https://doi.org/10.1039/c6em90004f

[6] Z. L. Botterell, N. Beaumont, T. Dorrington, M. Steinke, R. C. Thompson, and P. K. Lindeque, "Bioavailability and effects of microplastics on marine zooplankton: A review," *Environmental Pollution*, vol. 245, pp. 98-110, 2019, doi: https://doi.org/10.1016/j.envpol.2018.10.065

[7] R. Muñiz and M. S. Rahman, "Microplastics in coastal and marine environments: A critical issue of plastic pollution on marine organisms, seafood contaminations, and human health implications," *Journal of Hazardous Materials Advances*, vol. 18, pp. 1-22, 2025, doi: https://doi.org/10.1016/j.hazadv.2025.100663

[8] S. Tanasupawat, T. Takehana, S. Yoshida, K. Hiraga, and K. Oda, "Ideonella sakaiensis sp. nov., isolated from a microbial consortium that degrades poly(ethylene terephthalate)," *International Journal of Systematic and Evolutionary Microbiology*, vol. 66, no. 8, pp. 2813-2818, 2016, doi: https://doi.org/10.1099/ijsem.0.001058

[9] K. Kaur, S. Sharma, N. Shree, and R. Mehrotra, "Recent Advancements and Mechanism of Plastics Biodegradation Promoted by Bacteria: A Key for Sustainable Remediation for Plastic Wastes," *Biosciences Biotechnology Research Asia*, vol. 20, no. 1, pp. 1-12, 2023, doi: http://dx.doi.org/10.13005/bbra/3063

[10] L. Shi, H. Liu, S. Gao, Y. Weng, and L. Zhu, "Enhanced Extracellular Production of *Is*PETase in Escherichia coli via Engineering of the pelB Signal Peptide," *Journal of Agricultural and Food Chemistry*, vol. 69, no. 7, pp. 2245-2252, 2021, doi: https://doi.org/10.1021/acs.jafc.0c07469

[11] X. Huang, L. Cao, Z. Qin, S. Li, W. Kong, and Y. Liu, "Tat-Independent Secretion of Polyethylene Terephthalate Hydrolase PETase in *Bacillus subtilis* 168 Mediated by Its Native Signal Peptide," *Journal of Agricultural and Food Chemistry*, vol. 66, no. 50, pp. 13217-13227, 2018, doi: https://doi.org/10.1021/acs.jafc.8b05038

[12] D. Moog, J. Schmitt, J. Senger, J. Zarzycki, K. Rexer, U. Linne, T. Erb, and U. G. Maier, "Using a marine microalga as a chassis for polyethylene terephthalate (PET) degradation," *Microbial Cell Factories*, vol. 18, no. 1, pp. 1-15, 2019, doi: https://doi.org/10.1186/s12934-019-1220-z

[13] G. Di Rocco, H. N. Taunt, M. Berto, H. O. Jackson, D. Piccinini, A. Carletti, G. Scurani, N. Braidi, and S. Purton, "A PETase enzyme synthesised in the chloroplast of the microalga *Chlamydomonas reinhardtii* is active against post-consumer plastics," *Scientific Reports*, vol. 13, no. 1, pp. 1-10, 2023, doi: https://doi.org/10.1038/s41598-023-37227-5

[14] S. Joo, I. J. Cho, H. Seo, H. F. Son, H. Sagong, T. J. Shin, S. Y. Choi, S. Y. Lee, and K. Kim, "Structural insight into molecular mechanism of poly(ethylene terephthalate) degradation," *Nature Communications*, vol. 9, no. 1, pp. 1-12, 2018, doi: https://doi.org/10.1038/s41467-018-02881-1

[15] L. Niccolini, G. De Simone, M. Seggiani, N. Barbani, D. Rossi, M. Cappello, D. Levin, F. Spina, X. Yan, G. Petroni, I. Buttino, and S. Di Gregorio, "Nature-inspired biodegradation of poly(butylene succinate-co-adipate): the potential of *Cladosporium psychrotolerans* isolated from marine copepods," *Environmental Science and Pollution Research*, vol. 32, no. 53, pp. 29789-29802, 2025, doi: https://doi.org/10.1007/s11356-025-37262-x

[16] T. S. Ogunlusi, S. S. Ikoyo, M. Dadashipour, and H. Gao, "Engineering *Is*PETase and Its Homologues: Advances in Enzyme Discovery and Host Optimisation," *International Journal of Molecular Sciences*, vol. 26, no. 14, pp. 1-18, 2025, doi: https://doi.org/10.3390/ijms26146797

[17] W. Xu, "Microinjection and Micromanipulation: A Historical Perspective," *Methods in Molecular Biology*, pp. 1-16, 2018, doi: https://doi.org/10.1007/978-1-4939-8831-0_1

[18] Z. Z. Zakaria, S. Eisa-Beygi, F. M. Benslimane, R. Ramchandran, and H. C. Yalcin, "Design and Microinjection of Morpholino Antisense Oligonucleotides and mRNA into Zebrafish Embryos to Elucidate Specific Gene Function in Heart Development," *Journal of Visualized Experiments*, no. 186, pp. 1-26, 2022, doi: https://dx.doi.org/10.3791/63324